Scheme

25900801 Reg.No : .....
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE

EXAMINATION NOVEMBER 2025
FIRST SEMESTER

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer All Questions
Each Question carries 1 mark

1. c) The component drawing current
2. a) Directly proportional
3. a) Farad [K] / [CO1]
4. b) Base U] / [CO1]
5. b) Input voltage. [K] / [CO1]
6. c) Transformer K] / [CO2]
7. a) LDR [K] / [CO2]
8. b) Temperature control circuit's [K] / [CO2]
9. c) R1 × R2 / (R1 + R2) [K] / [CO1]
10. b) By magnetic induction [K] / [CO3]

Part B
Short answer Type Questions
Answer any 4 Questions
Each Question carries 5 marks

11. Explain forward current and reverse current in a diode. [U] / [CO1]

A diode conducts current only when it is forward biased.

- **Forward current:** When the anode is positive w.r.t the cathode, the depletion layer reduces and current flows easily.
- **Reverse current:** When the diode is reverse biased, only a very tiny leakage current flows because the depletion region widens. If reverse voltage is too high, breakdown may occur.

12. Extract the benefits of biometric authentication. [K] / [CO1]

- Provides **high security** because traits like fingerprint/iris cannot be easily copied.
- **Convenient**—no need to remember passwords or carry ID cards.
- Ensures **accurate user identification**.
- Harder to forge or hack compared to passwords.
- Fast verification and improves overall system integrity.

13. What are the main characteristics of P-type semiconductors? [U] / [CO1]

- Formed by **doping** pure silicon with **trivalent atoms** (Boron).
- Majority carriers: **Holes**.
- Minority carriers: **Electrons**.
- Possess **positive charge carriers** leading to better hole conduction.
- Used in forming **PN junctions**, diodes, and transistors.

14. In a clamper circuit, what role does the capacitor play? [U] / [CO2]

- The capacitor stores charge during part of the input cycle.
- It **shifts the entire waveform** up or down without changing its shape.
- Helps maintain a DC level across output (adds/subtracts DC bias).
- Works with the diode to clamp the signal at a chosen reference level.

15. What do you understand by avalanche breakdown? [K] / [CO1]

Avalanche breakdown occurs in a diode when **high reverse voltage** accelerates minority carriers so fast that they collide with atoms and create more charge carriers.
This multiplication of carriers causes a sudden **large reverse current**.
Common in Zener diodes above **6 V**.

16. How does a fuse provide overcurrent protection? [U] / [CO3]

- A fuse contains a **thin metal strip** with low melting point.
- When excessive current flows, the strip **heats up and melts**.
- This breaks the circuit and prevents damage to wiring or appliances.
- It provides simple, economical, and effective protection.

Part C
Essay Type Questions
Answer any 2 Questions
Each Question carries 10 marks

17. Explain the basic principle of resistors and capacitors. Briefly explain their classification. [U] / [CO1]

Resistors – Basic Principle

- A resistor opposes the flow of electric current.
- Based on Ohm's Law:
  $V=IRV=IR$
- Converts electrical energy into heat.
- Used for current limiting, voltage division, and biasing circuits.

Classification of Resistors

Fixed Resistors

- Carbon composition
- Metal film
- Wire-wound

Variable Resistors

- Potentiometers
- Rheostats

Special Resistors

- LDR (Light Dependent Resistor)
- Thermistors (NTC, PTC)

Capacitors – Basic Principle

- A capacitor stores electrical energy in an electric field.
- Made of two plates separated by a dielectric.
- Capacitive reactance:
  $Xc=12\pi fCXc=2\pi fC1$
- Used for filtering, coupling, bypassing, and timing circuits.

Classification of Capacitors

Fixed Capacitors

- Ceramic
- Electrolytic
- Film

Variable Capacitors

- Trimmer capacitors

Polarized & Non-Polarized Types

18. Explain the V-I Characteristics of JFET. [U] / [CO2]

Introduction

A JFET (Junction Field Effect Transistor) is a voltage-controlled device where current flows through a semiconductor channel.

Regions of V-I Characteristics

Ohmic (Linear) Region

- For small VDSVDS, the drain current IDID increases linearly with voltage.
- JFET behaves like a variable resistor.

Saturation (Pinch-off) Region

- After a certain voltage (pinch-off), current becomes constant.
- Controlled by gate voltage VGSVGS.

Cut-off Region

- If VGSVGS becomes more negative than **cut-off voltage VGS(off)VGS(off)**, ID=0ID=0

Effect of Gate Voltage

- More negative VGSVGS reduces channel width.
- Drain current equation:
  $ID=IDSS(1-VGSVGS(off))2ID=IDSS(1-VGS(off)VGS)2$

19. Explain the working principle of a half-wave rectifier and discuss its limitations and applications. [U] / [CO2]

Working Principle

- Uses a single diode.
- During **positive half cycle**, diode becomes forward biased → passes current → output obtained.
- During **negative half cycle**, diode becomes reverse biased → current blocked → no output.

Waveform Behaviour

- Output is pulsating DC containing ripples.

Limitations

- Low efficiency (40.6%).
- High ripple factor (1.21).
- Not suitable for high-power applications.
- Utilizes only half of AC cycle.

Applications

- Signal demodulation (AM radios).
- Low-power DC supplies.
- Voltage multiplier circuits.
- Battery charging (simple circuits).

20. What are the advantages of using a DPST switch over an SPST switch?

SPST (Single Pole Single Throw) **K/ [CO3]**

- Controls **only one** circuit line.
- Turns ON/OFF a single conductor.

DPST (Double Pole Single Throw)

- Controls **two separate lines simultaneously** with one switch.

Advantages of DPST Over SPST

Better safety

- Disconnects both phase and neutral lines at the same time.

Controls two circuits together

- Useful for appliances requiring dual isolation.

Reduces risk of electric shock

- Full disconnection avoids accidental live contact.

Used in high-power devices

- Ensures complete cut off during servicing.

More reliable operation

- Prevents leakage or back feed from the second line.